

- A least squares solution to $A\mathbf{x} = \mathbf{b}$ is a vector $\hat{\mathbf{x}}$ that makes $A\hat{\mathbf{x}}$ as close to $\mathbf{b}$ as possible.
- A least squares solution to $A\mathbf{x} = \mathbf{b}$ is a vector $\hat{\mathbf{x}}$ that minimizes the distance $\|\mathbf{b} - A\mathbf{x}\|$, known as the least squares error.

How to Solve

2 Using the Normal Equations

Theorem 13: The Normal Equations

The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ coincides with the set of solutions of the *normal equations*: $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$. The normal equations are always consistent.

To find a least squares solution using the normal equations, compute $A^T A$ and $A^T \mathbf{b}$, then solve the new system $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$. Each solution will be a least squares solution $\hat{\mathbf{x}}$ to $A\mathbf{x} = \mathbf{b}$.

(<https://courses.math.tufts.edu/math70/Section%20Summaries/Chapter6/sect%206.5.pdf>)

Additional Exercises Lab9-10

Ex12

The following data shows the sales (in million dollars) of a company.

x	2015	2016	2017	2018	2019
y	12	19	29	37	45

Can you estimate the sales in the year 2020 using a least squares line?

Ex13

Find the vector in the column space of A closest to the vector b where:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -3 \\ 2 & 6 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Hint: a vector in the column space of A should have the following format: $x \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}$. Thus, using least square solution to find x and y coefficients.